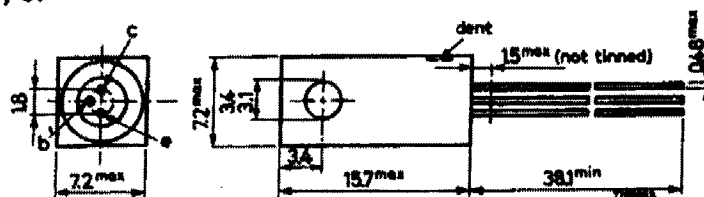


AC187 / AC187K/ AC187/01/ AC188

MECHANICAL DATA (continued)

Dimensions in mm

AC187/01



The dent indicates the collector

RATINGS Limiting values in accordance with the Absolute Maximum System (IEC 134)

Voltages

Collector-base voltage (open emitter)	V_{CBO}	max.	25 V
Collector-emitter voltage (open base)	V_{CEO}	max.	15 V
Collector-emitter voltage $I_C \leq 600 \text{ mA}$; $R_{BE} \leq 1 \Omega$	V_{CER}	max.	18 V
Emitter-base voltage (open collector)	V_{EBO}	max.	10 V

Currents

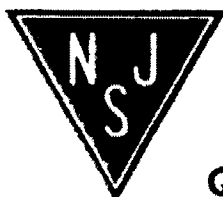
Collector current (d.c. or average over any 50 ms period)	I_C	max.	1 A
Collector current (peak value)	I_{CM}	max.	2 A

Power dissipation

Total power dissipation up to $T_{amb} = 35^\circ\text{C}$	P_{tot}	max.	1.0 W
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Temperatures

Storage temperature	T_{stg}	-55 to +75	$^\circ\text{C}$
Junction temperature	T_j	max.	90 $^\circ\text{C}$

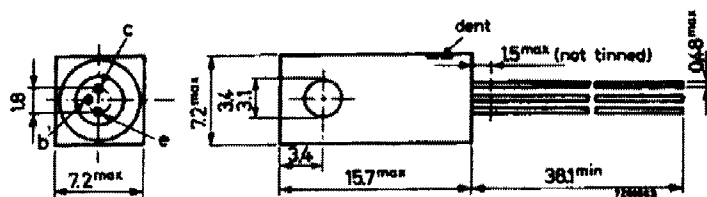


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Quality Semi-Conductors

MECHANICAL DATA (continued) **AC187/01**

Dimensions in mm



The dent indicates the collector

RATINGS Limiting values in accordance with the Absolute Maximum System (IEC 134)

Voltages

Collector-base voltage (open emitter)	V_{CBO}	max.	25 V
Collector-emitter voltage (open base)	V_{CEO}	max.	15 V
Collector-emitter voltage $I_C \leq 600 \text{ mA}$; $R_{BE} \leq 1 \Omega$	V_{CER}	max.	18 V
Emitter-base voltage (open collector)	V_{EBO}	max.	10 V

Currents

Collector current (d.c. or average over any 50 ms period)	I_C	max.	1 A
Collector current (peak value)	I_{CM}	max.	2 A

Power dissipation

Total power dissipation up to $T_{amb} = 35^\circ\text{C}$	P_{tot}	max.	1.0 W
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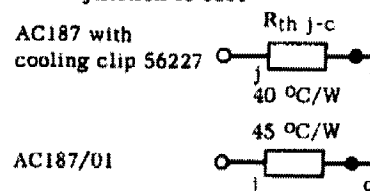
Temperatures

Storage temperature	T_{stg}	-55 to +75	$^\circ\text{C}$
Junction temperature	T_j	max.	90 $^\circ\text{C}$

THERMAL RESISTANCE

From junction to ambient in free air
 without cooling clip
 with cooling clip 56227
 with cooling clip 56227 on
 1.5 mm blackened Al. heatsink
 with cooling clip 56227 on infinit

From junction to case



CHARACTERISTICS

Collector cut-off current

$I_E = 0$; $V_{CB} = 25 \text{ V}$

$I_E = 0$; $V_{CB} = 25 \text{ V}$; $T_j = 90^\circ\text{C}$

$-V_{BE} = 1.0 \text{ V}$; $V_{CE} = 25 \text{ V}$

Emitter cut-off current

$I_C = 0$; $V_{EB} = 10 \text{ V}$

$I_C = 0$; $V_{EB} = 10 \text{ V}$; $T_j = 90^\circ\text{C}$

Base-emitter voltage

$I_C = 5 \text{ mA}$; $V_{CE} = 10 \text{ V}$

$I_C = 300 \text{ mA}$; $V_{CE} = 1 \text{ V}$

Emitter-base floating voltage

$I_E = 0$; $V_{CB} = 25 \text{ V}$; $T_j = 90^\circ\text{C}$

THERMAL RESISTANCE

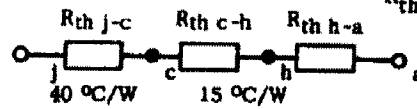
From junction to ambient in free air

without cooling clip
with cooling clip 56227
with cooling clip 56227 on
1.5mm blackened Al. heatsink of 12.5 cm²
with cooling clip 56227 on infinite heatsink

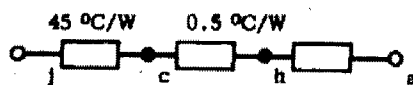
	AC187	AC187/01
$R_{th\ j-a}$ = 290	180 °C/W	
$R_{th\ j-a}$ = 140		°C/W
$R_{th\ j-a}$ = 80	70.5 °C/W	
$R_{th\ j-a}$ = 55		°C/W
$R_{th\ j-c}$ = 40	45 °C/W	

From junction to case

AC187 with
cooling clip 56227



AC187/01



CHARACTERISTICS

$T_j = 25\text{ °C}$ unless otherwise specified

Collector cut-off current

$I_E = 0$; $V_{CB} = 25\text{ V}$

I_{CBO}	typ. 15 μA
	< 100 μA

$I_E = 0$; $V_{CB} = 25\text{ V}$; $T_j = 90\text{ °C}$

I_{CBO}	< 2.5 mA
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$-V_{BE} = 1.0\text{ V}$; $V_{CE} = 25\text{ V}$

I_{CEX}	< 100 μA
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Emitter cut-off current

$I_C = 0$; $V_{EB} = 10\text{ V}$

I_{EBO}	typ. 15 μA
	< 100 μA

$I_C = 0$; $V_{EB} = 10\text{ V}$; $T_j = 90\text{ °C}$

I_{EBO}	typ. 1.2 mA
	< 2.5 mA

Base-emitter voltage

$I_C = 5\text{ mA}$; $V_{CE} = 10\text{ V}$

V_{BE}	95 to 135 mV
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$I_C = 300\text{ mA}$; $V_{CE} = 1\text{ V}$

V_{BE}	< 550 mV
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Emitter-base floating voltage

$I_E = 0$; $V_{CB} = 25\text{ V}$; $T_j = 90\text{ °C}$

V_{EBf}	< 400 mV
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